



SUMMARY REPORT



Customer Ride Data Preprocessing & Feature Engineering Pipeline



Project Objective

The primary objective of this project was to preprocess and transform ride-sharing data collected from multiple heterogeneous sources, including Excel, JSON, and SQL databases. The project aimed to clean raw data, handle missing values, detect and treat outliers, perform feature engineering, and generate a structured, machine-learning-ready dataset suitable for predictive analytics and business intelligence applications.



Datasets Used

The project utilized the following datasets:



riders.xlsx

Contains rider-related information such as:

- Rider ID
- Age
- Gender
- Registration Details
- Rider Profile Attributes



trips.json

Contains trip-related information including:

- Trip ID
- Rider ID
- Distance Travelled
- Trip Duration
- Fare Amount
- Payment Mode
- Ride Date and Time
- Surge Pricing Indicator

city_zones.sql

Contains city zone information such as:

- Zone Name
 - Population Density
 - Traffic Index
 - Average Speed
 - Zone Type
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Data Cleaning & Preprocessing

Several preprocessing techniques were applied to improve data quality and consistency:

Duplicate Record Removal

- Identified and removed duplicate records from all datasets.

Missing Value Treatment

- Numerical missing values were handled using **Mean Imputation**.
- Categorical missing values were filled using **Most Frequent Imputation**.
- Advanced missing value treatment was performed using **KNN Imputer** for selected numerical features.

Invalid Data Handling

- Removed records containing invalid fare amounts, distances, or inconsistent values.
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Outlier Detection & Treatment

To improve data reliability and model performance, multiple outlier handling techniques were applied:

♦ Z-Score Method

Applied on:

- Fare Amount
- Distance Travelled

♦ IQR (Interquartile Range) Method

Applied on:

- Trip Duration

◆ Winsorization

Used where necessary to reduce the impact of extreme observations without removing valuable records.

Data Transformation

The following transformation techniques were applied:

Datetime Feature Extraction

Ride date and time were converted into:

- Ride Hour
- Ride Day
- Ride Month

Encoding Techniques

- Label Encoding
- One-Hot Encoding

Mathematical Transformations

- Log Transformation
- Square Root Transformation

These transformations helped improve feature distributions and model readiness.

Feature Engineering

Several meaningful features were created to enhance analytical capabilities and predictive performance:

Created Features

- Average Ride Distance
- Average Ride Fare
- Fare Per Kilometer
- Peak Hour Indicator (is_peak_hour)
- Days Since Rider Signup
- Ride Cancellation Rate
- Surge Pricing Flag
- Traffic-Based Zone Features
- Time-Based Ride Features

These engineered variables provide additional business insights and improve machine learning model effectiveness.

Feature Scaling

To standardize feature values and improve model performance, the following scaling techniques were applied:

- **StandardScaler**
- **MinMaxScaler**

This ensured that numerical variables were transformed into comparable scales.

Final Output

The final cleaned and transformed dataset was successfully generated and exported as:

 **final_prepared_rides_dataset.csv**

The dataset is fully prepared for:

- Machine Learning Applications
 - Predictive Analytics
 - Customer Behavior Analysis
 - Ride Demand Forecasting
 - Fare Prediction Models
 - Business Intelligence Reporting
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Conclusion

This project successfully demonstrated a complete end-to-end data preprocessing and feature engineering workflow using ride-sharing datasets from multiple sources. Through data cleaning, missing value treatment, outlier handling, transformation, encoding, scaling, and feature engineering, the raw data was converted into a high-quality machine-learning-ready dataset.

The final dataset provides a strong foundation for advanced analytics, predictive modeling, and data-driven decision-making in ride-sharing and transportation systems.

 **Prepared By:**

Misari Dhorajiya

Data Science & Machine Learning Enthusiast